

Deploying a Static Website on AWS

S3 + CloudFront + Route 53 + ACM (SSL)

Project: saurabhchavan.shop

This document walks through the steps taken to host a static portfolio website on AWS using Amazon S3 for storage, AWS Certificate Manager (ACM) for SSL/TLS, Amazon CloudFront as the content delivery network (CDN), and Amazon Route 53 for DNS management — resulting in a fully secure, custom-domain website served over HTTPS.

Step 1: Create an S3 Bucket

- Navigated to Amazon S3 → Create bucket.
- Selected **General purpose** bucket type and **Global namespace**.
- Named the bucket **saurabhchavan.shop** (matching the custom domain name).
- Kept default settings (ACLs disabled — recommended) and created the bucket.

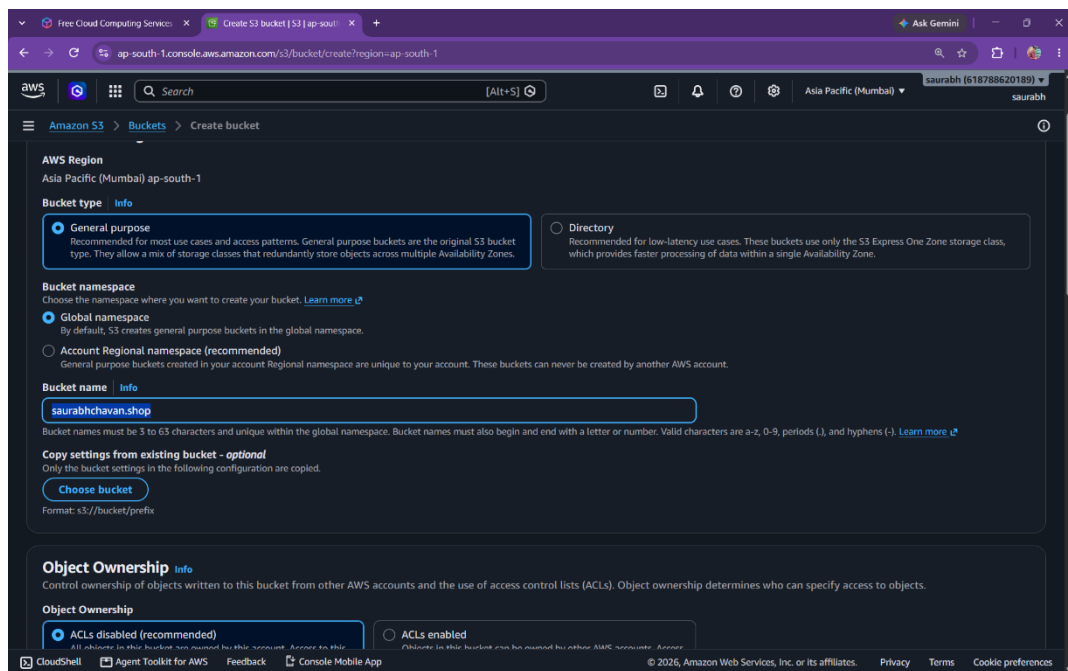


Fig 1: Creating the S3 bucket 'saurabhchavan.shop' in the Asia Pacific (Mumbai) region.

Step 2: Upload Website Files

- Opened the newly created bucket.
- Uploaded the static website files: **index.html**, **style.css**, **script.js**, and an **assets/** folder containing images.

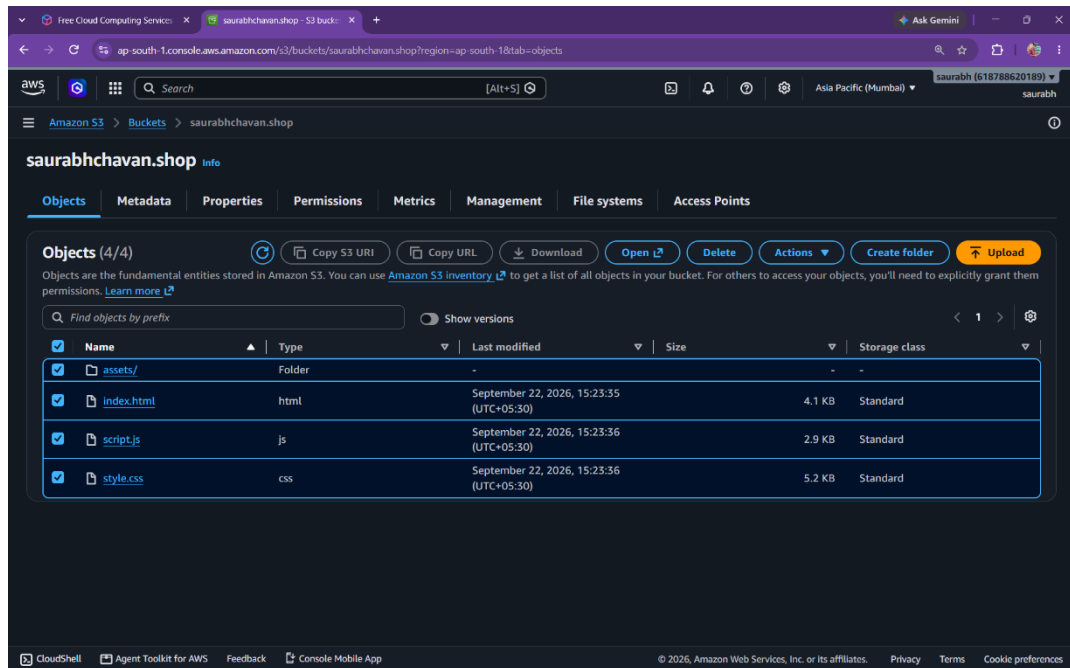


Fig 2: Website objects uploaded to the S3 bucket (4/4 objects).

Step 3: Request an SSL/TLS Certificate (AWS Certificate Manager)

- Navigated to AWS Certificate Manager (ACM) — certificate created in the **US East (N. Virginia)** region, since CloudFront requires certificates from this region.
- Requested a public certificate for the domain **saurabhchavan.shop**.
- Validated domain ownership and confirmed the certificate status as **Issued**.

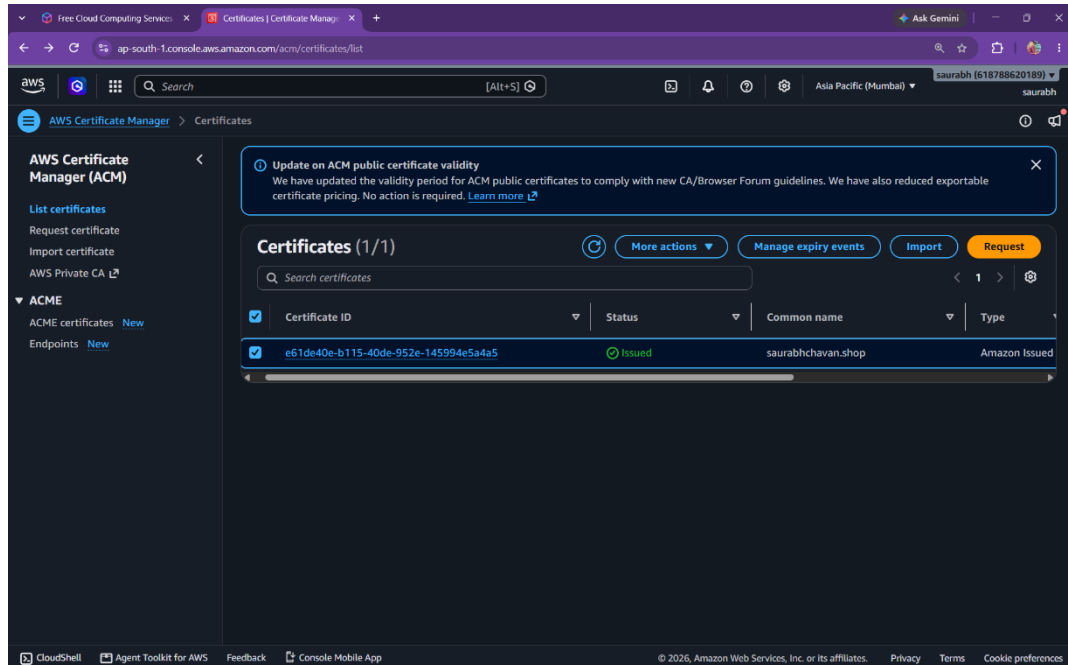


Fig 3: ACM certificate for saurabhchavan.shop showing status 'Issued'.

Step 4: Create a CloudFront Distribution

- Went to CloudFront → Create distribution.
- Chose **Single website configuration**.
- Entered the domain **saurabhchavan.shop** — CloudFront auto-detected the matching Route 53 hosted zone.
- Selected the S3 bucket as the distribution's **origin**.

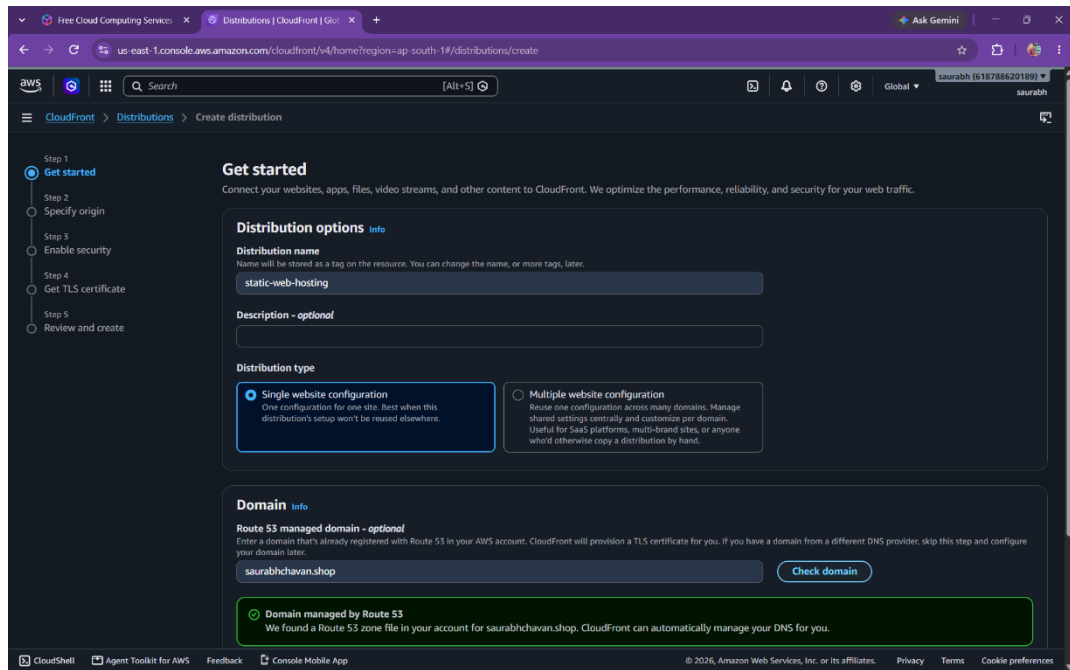


Fig 4: Creating the CloudFront distribution with the domain linked to Route 53.

Step 5: Configure Custom SSL Certificate and Root Object

- Attached the custom SSL certificate issued from ACM to the distribution.
- Set the security policy to **TLSv1.2_2021 (recommended)**.
- Enabled **HTTP/2** support.
- Set the **Default root object** to **index.html**.

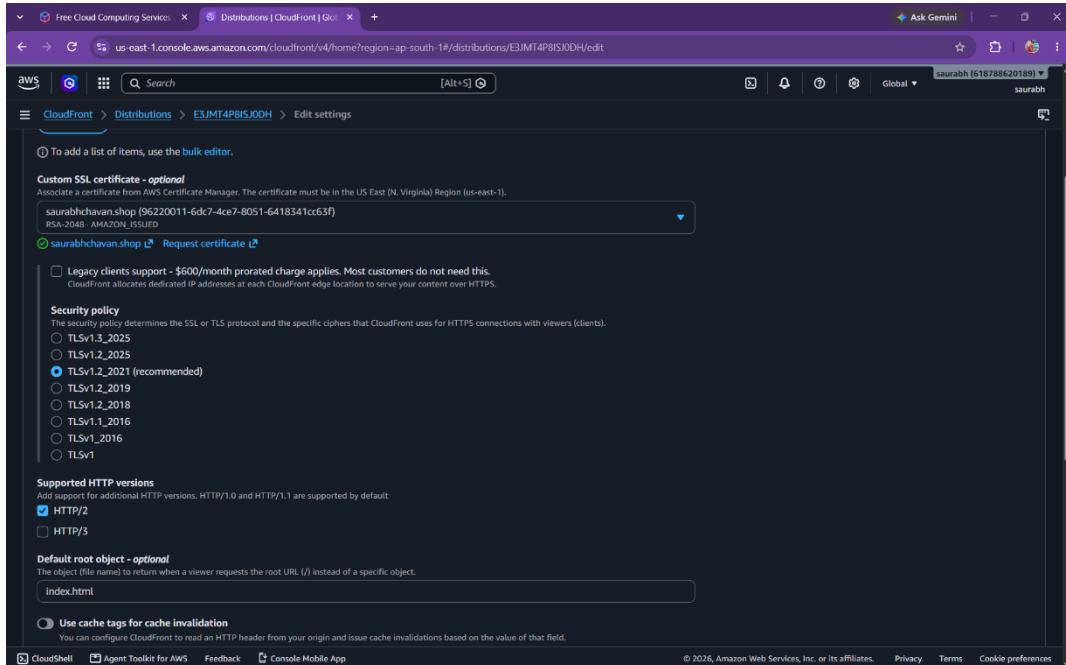


Fig 5: CloudFront distribution settings — custom SSL certificate and default root object.

Step 6: Verify Origins

- Confirmed the origin correctly points to the S3 bucket (**saurabhchavan.shop.s3...**).
- Origin type verified as **S3**.

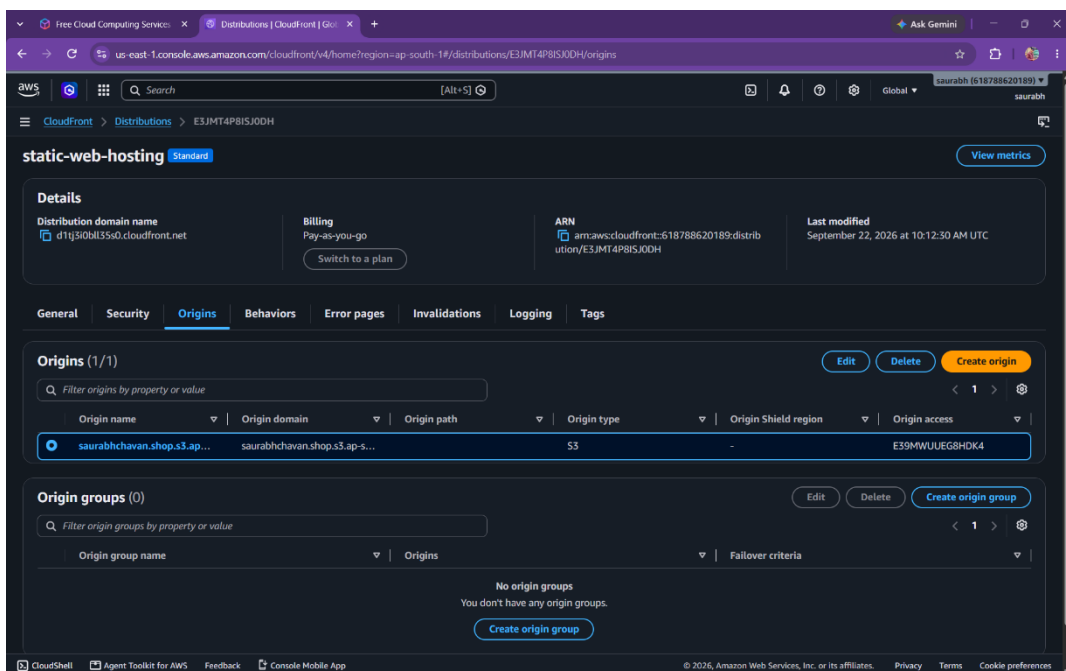


Fig 6: Origins tab confirming the S3 bucket as the distribution's origin.

Step 7: Review the Distribution Dashboard

- Checked the distribution's **General** tab.
- Confirmed the CloudFront domain name (e.g., **d1tj3i0bil35s0.cloudfront.net**), the alternate domain name (**saaurabhchavan.shop**), attached SSL certificate, and default root object.

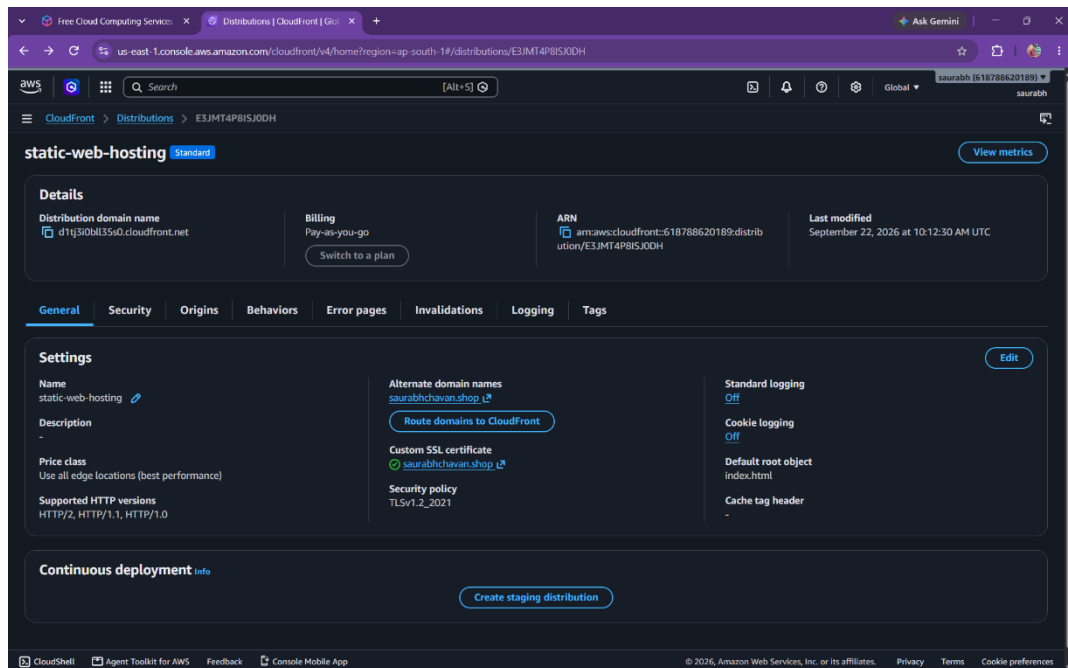


Fig 7: CloudFront distribution dashboard summary ('static-web-hosting').

Step 8: Create a Hosted Zone in Route 53

- Went to Route 53 → Hosted zones → Create hosted zone.
- Created a **public hosted zone** for **saaurabhchavan.shop**.

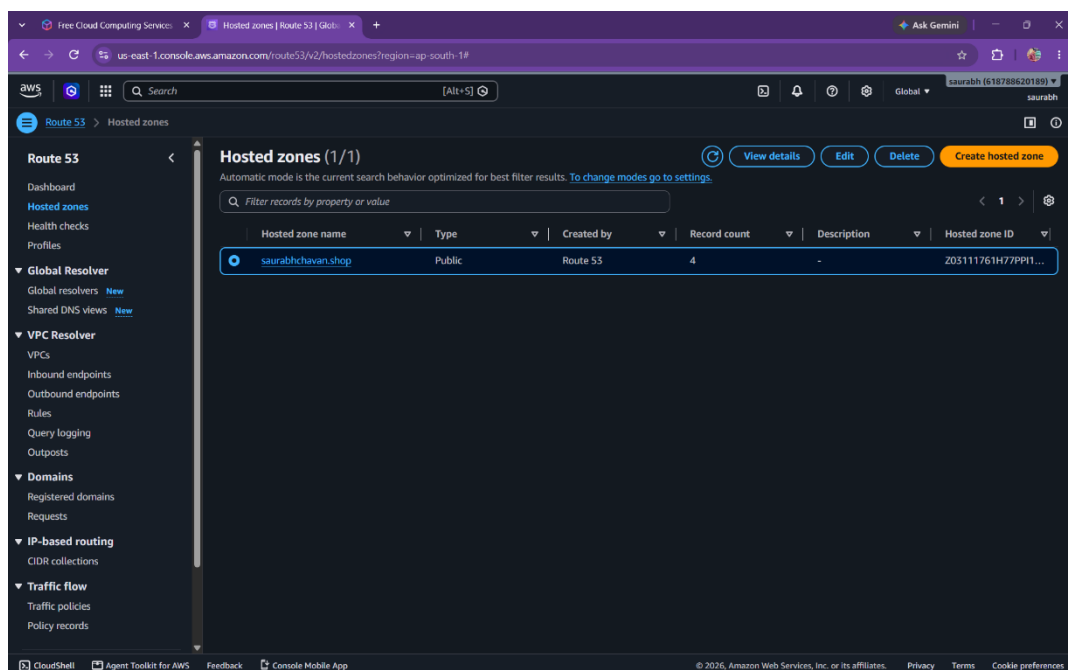


Fig 8: Route 53 hosted zone created for saaurabhchavan.shop.

Step 9: Edit DNS Record — Alias to CloudFront

- Selected the **A record** for the domain.
- Enabled **Alias**.
- Set "Route traffic to" → **Alias to CloudFront distribution**.
- Selected the CloudFront distribution created earlier.

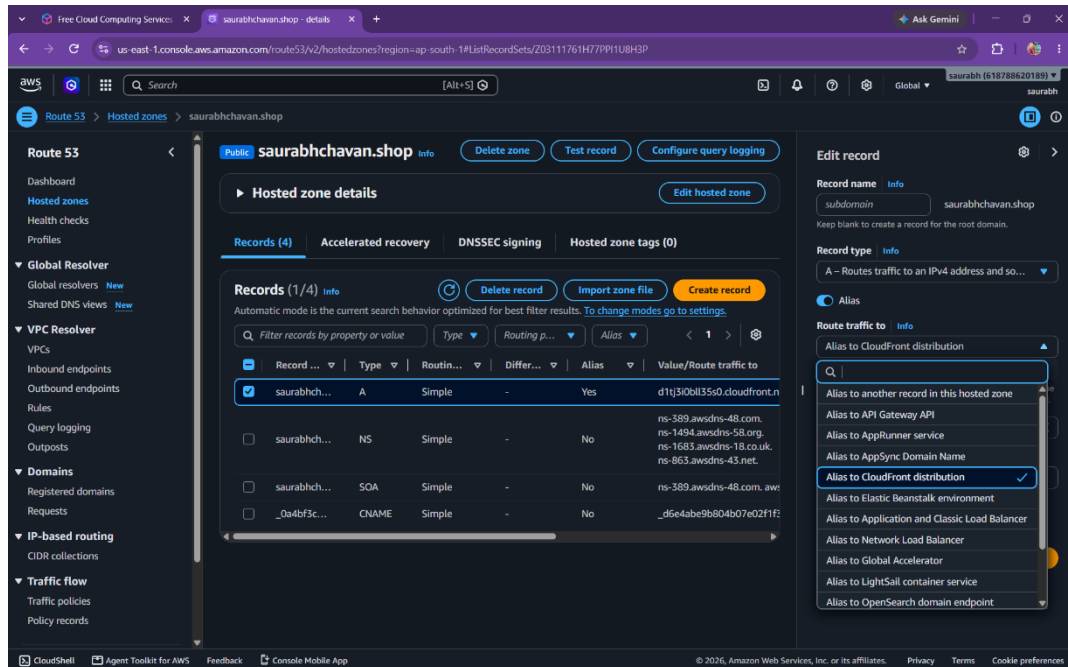


Fig 9: Editing the Route 53 A record to alias traffic to the CloudFront distribution.

Step 10: Verify the Live, Secure Website

- Opened **saurabhchavan.shop** in the browser.
- Confirmed the padlock icon shows **"Connection is secure"** — confirming HTTPS is working correctly via ACM + CloudFront.
- The portfolio website loaded successfully on the custom domain.

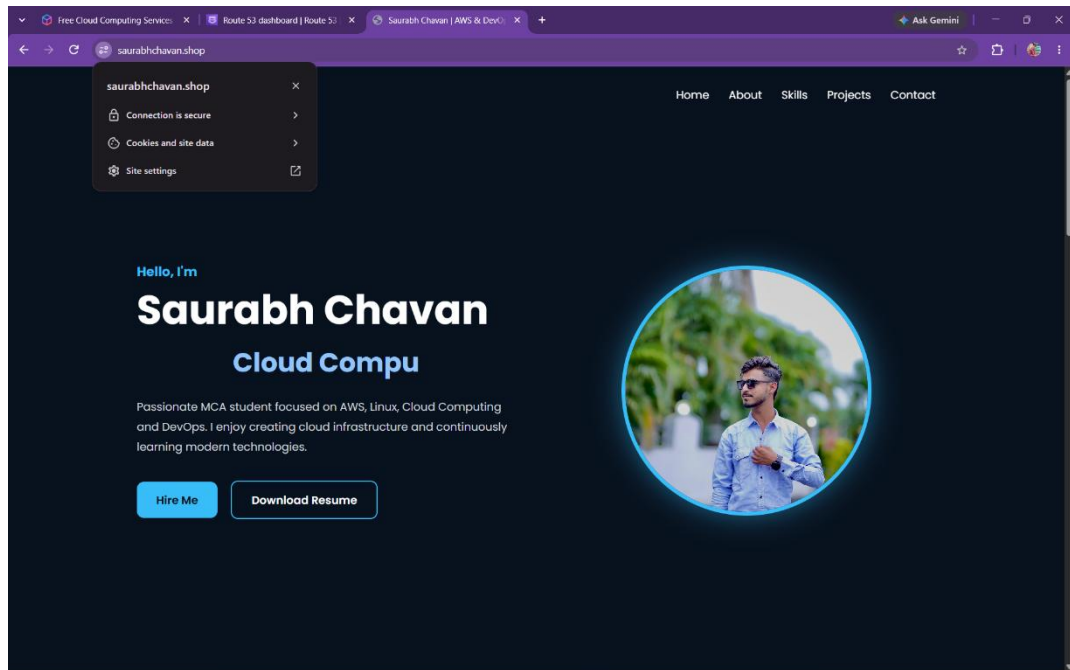


Fig 10: Live website at saurabhchavan.shop with a secure HTTPS connection.

Architecture Summary

The final architecture flows as follows: a visitor's browser resolves **saurabhchavan.shop** via a Route 53 alias record, which points to the CloudFront distribution. CloudFront terminates HTTPS using the ACM-issued certificate and serves cached static content from the S3 bucket origin, delivering the website securely and with low latency from AWS edge locations worldwide.

Traffic flow: Route 53 (DNS) → CloudFront (CDN + HTTPS via ACM certificate) → S3 Bucket (static file origin)